

Does the term premium pay for the duration?

Evidence from Polish sovereigns.

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Motivation

EX-ANTE PREMIA VERSUS REALISED ATTRIBUTION

- On the Polish curve the ex-ante term premium is positive on average: roughly +105 bp at the 10-year tenor under ACM with the BRW small-sample bias correction, and positive on the large majority of days. The textbook implication is that a duration position over rolling short-term cash earns this premium.
- The empirical question is the realised counterpart: of the 4.14% compound annual return the published TBSP index delivered over nineteen years, **how much is attributable to term-premium realisation** rather than to coupon accrual, reinvestment and roll-down?
- The benchmark cash rate throughout is the NBP reference rate, a proxy for the upper envelope of retail deposit rates.

A seven-component identity for the quarterly index return

DECOMPOSITION

Each quarter's TBSP log-return is partitioned in closed form:

$$r_t = \underbrace{CC_t + RP_t + RD_t}_{\text{CarryRoll}} + \underbrace{ERC_t + TPC_t}_{\text{CurveMove}} + Cnvx_t + Res_t$$

- CC coupon accrual, RP reinvestment proceeds, RD clean-price roll-down along an unchanged curve: the return to an unchanged state of the world.
- ERC and TPC : revaluation from the expected-rate and term-premium legs of the ACM–BRW decomposition of the fitted zero curve.
- $Cnvx$ second-order convexity; Res the closure residual. The sum reproduces the published index by construction: the identity is an attribution, not a model.
- The split of CurveMove rests on the fitted-curve decomposition $y_t(\tau) = y_t^{rf}(\tau) + tp_t(\tau)$:

| Data and replication quality

THE PANEL

- Published Treasury BondSpot Poland (TBSP) index, 2006-12-29 to 2026-05-15, 80 quarter-end snapshots, replicated bond by bond on an 89-bond BondSpot panel with daily LW-NSS zero curves.
- Closure residual: +20 bp cumulative over the full sample. Tracking gap against the published index: -106 bp cumulative (-5.4 bp per year). Both are an order of magnitude below the components being attributed.

| Result 1: the carry-and-roll bloc

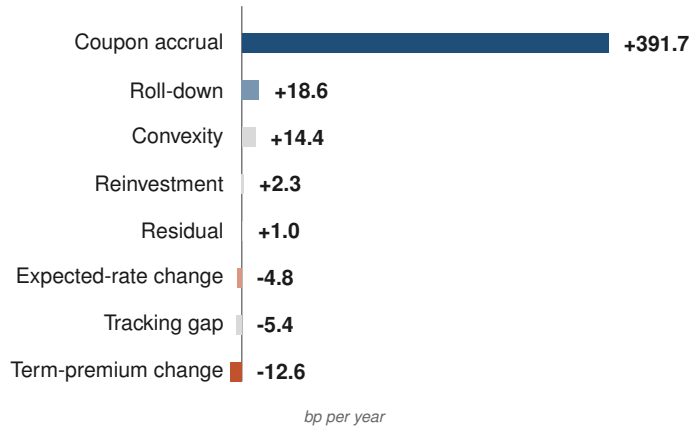
CUMULATIVE ATTRIBUTION

+413 bp/yr

cumulative CarryRoll contribution ($CC + RP + RD$), +8,010 bp over the sample, against +7,866 bp delivered by the published index. The income bloc alone accounts for essentially the whole realised return.

The full attribution, per year of sample

CUMULATIVE CONTRIBUTION / SAMPLE LENGTH



Result 2: full-sample inference on the curve-move bloc

HAC TESTS AND BOOTSTRAP INTERVALS

- *ERC* + *TPC* cumulates to -337 bp over the sample (-17 bp per year), largely the unreversed portion of the 2022 tightening-cycle revaluation.
- The full-sample tests are uninformative: the mean realised *TPC* of -3.1 bp per quarter has HAC $p = 0.83$, with a bootstrap confidence interval on the cumulative of $[-2,534, +1,776]$ bp; the TBSP-versus-cash excess of $+17.6$ bp per quarter has $p = 0.55$. Both intervals bracket zero.
- The tests: with $e_t = r_t^{TBSP} - r_t^{cash}$ the null is $H_0: \mathbb{E}[e_t] = 0$, assessed by $t = \bar{e} / \sqrt{\hat{V}_{NW}/T}$ and by bootstrap intervals on cumulative sums; the same construction applies to the *TPC* series.
- The reason for the wide intervals is not noise throughout the sample but the concentration of the variance in a single regime episode.

| The 2021Q4–2022Q3 tightening episode

ONE EPISODE DOMINATES THE SAMPLE MOMENTS

- Four quarters contribute –962 bp of cumulative term-premium change: four times the full sample's net –245 bp.
- The index fell from 2,175 to 1,703 peak-to-trough, a –21.7% drawdown, only partially reversed after 2023.
- Treating these quarters as part of a homogeneous sample conflates a rare regime realisation with the unconditional distribution; the paper conditions on the episode explicitly rather than through dummies.

Result 3: the conditional excess return

EXCLUDING THE FOUR EPISODE QUARTERS

+46.5 bp/q

mean TBSP-versus-rolling-cash excess on the 74-quarter sub-sample, HAC $p = 0.003$, cumulative bootstrap confidence interval $[+1,217, +5,534]$ bp. The excess is positive on 74 of the 78 quarter-ends in the full sample.

| Interpretation

RECONCILING EX-ANTE AND REALISED PREMIA

- A positive ex-ante term premium and a realised term-premium contribution indistinguishable from zero are not contradictory: realisation risk is concentrated in rare, persistent regime episodes, so unconditional sample means converge slowly.
- A TBSP-tracking exposure is characterised more accurately as a carry position between the term spread and the policy rate, bearing low-frequency regime risk, than as a systematic harvest of the ex-ante premium.
- The characterisation reproduces the observed moments: the long-run excess over cash, the modest trailing information ratio (0.21), and the depth of the 2022 drawdown.

| Data and econometrics

SPECIFICATION SUMMARY

- **Identity.** Closed-form seven-component partition of the quarterly log-return of a coupon-bearing bond, aggregated with published TBSP weights; derivation and closure properties in the paper's appendix.
- **Curves.** Daily LW-NSS zero curves; expected-rate and term-premium legs from ACM with the BRW small-sample bias correction.
- **Inference.** Newey–West (HAC) tests on quarterly means and bootstrap confidence intervals on cumulative contributions; the regime episode is handled by explicit conditioning.
- **Verification.** The closure residual and the tracking gap against the published index are reported, not absorbed: +20 bp and –106 bp cumulative respectively.

Companion data and replication

YIELDCARTOGRAPHY.COM

- The identity runs on each quarter as it closes: index-level cards and a per-bond explorer at yieldcartography.com/total-return.
- The ACM–BRW expected-rate and term-premium legs update daily on [/term-premia/](https://yieldcartography.com/term-premia/); the current quarter-to-date decomposition is carried on the cockpit.
- Underlying CSVs are downloadable from the dashboards.

Thank you.

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yieldcartography.com/slides/duration/